

VIRTUAL SCREENING I

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- Descriptor Based Screening (Lipinski, Verbers etc.)
- Similarity and Similarity Search
- Pharmacophore Screening
- Fragment Based Screening

Virtual Screening

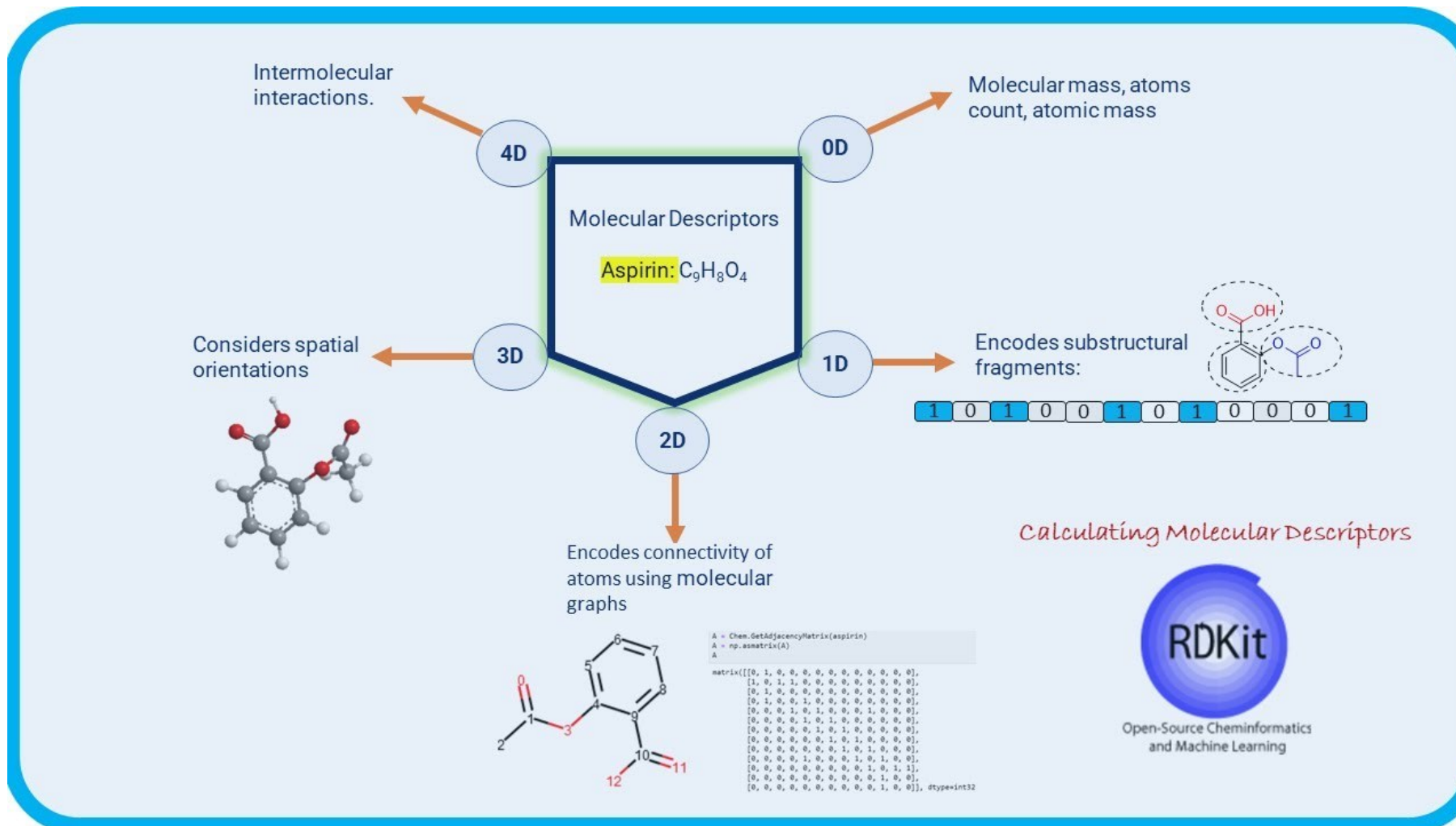


Virtual screening refers to methods necessary to **select compounds of interest** for further experimentation

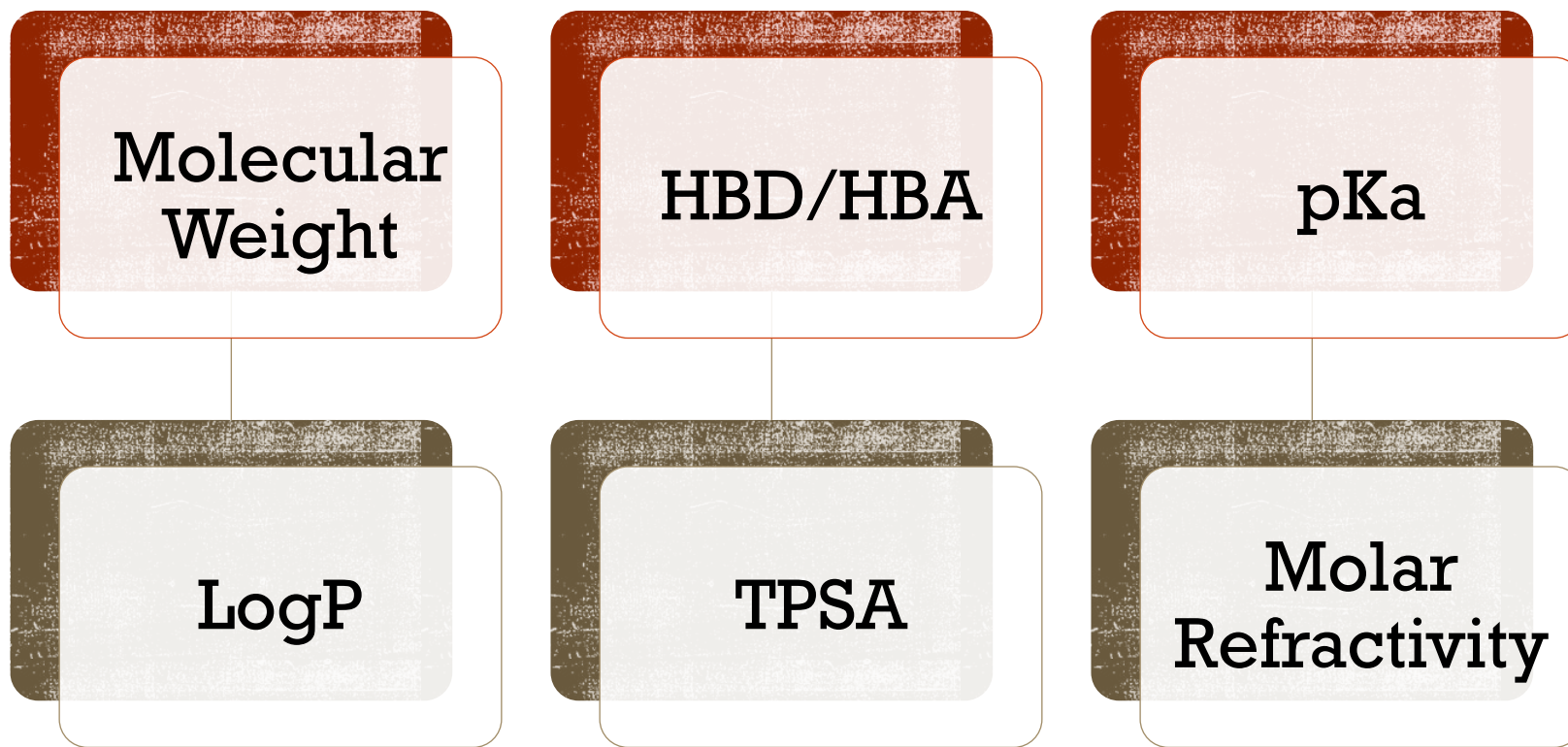
These interest might include:

- **A given physicochemical parameter**
- **Activity**
- **Pharmacokinetic parameters**
- **Binding**

Chemical/Molecular Descriptors



PHYSICOCHEMICAL DESCRIPTORS



Implementation in RDKit

- Most Basic representation of molecules
- Encodes real physical information
- Simplest is the Lipinski descriptors (as we will see shortly)
- RDKit has about 200+ of such descriptors including aromatic rings etc

```
#Import the library
```

```
from rdkit import Chem  
from rdkit.Chem import Descriptors
```

```
# Create a molecule object for Aspirin (C9H8O4)  
mol = Chem.MolFromSmiles('CC(=O)Oc1ccccc1C(=O)O')
```

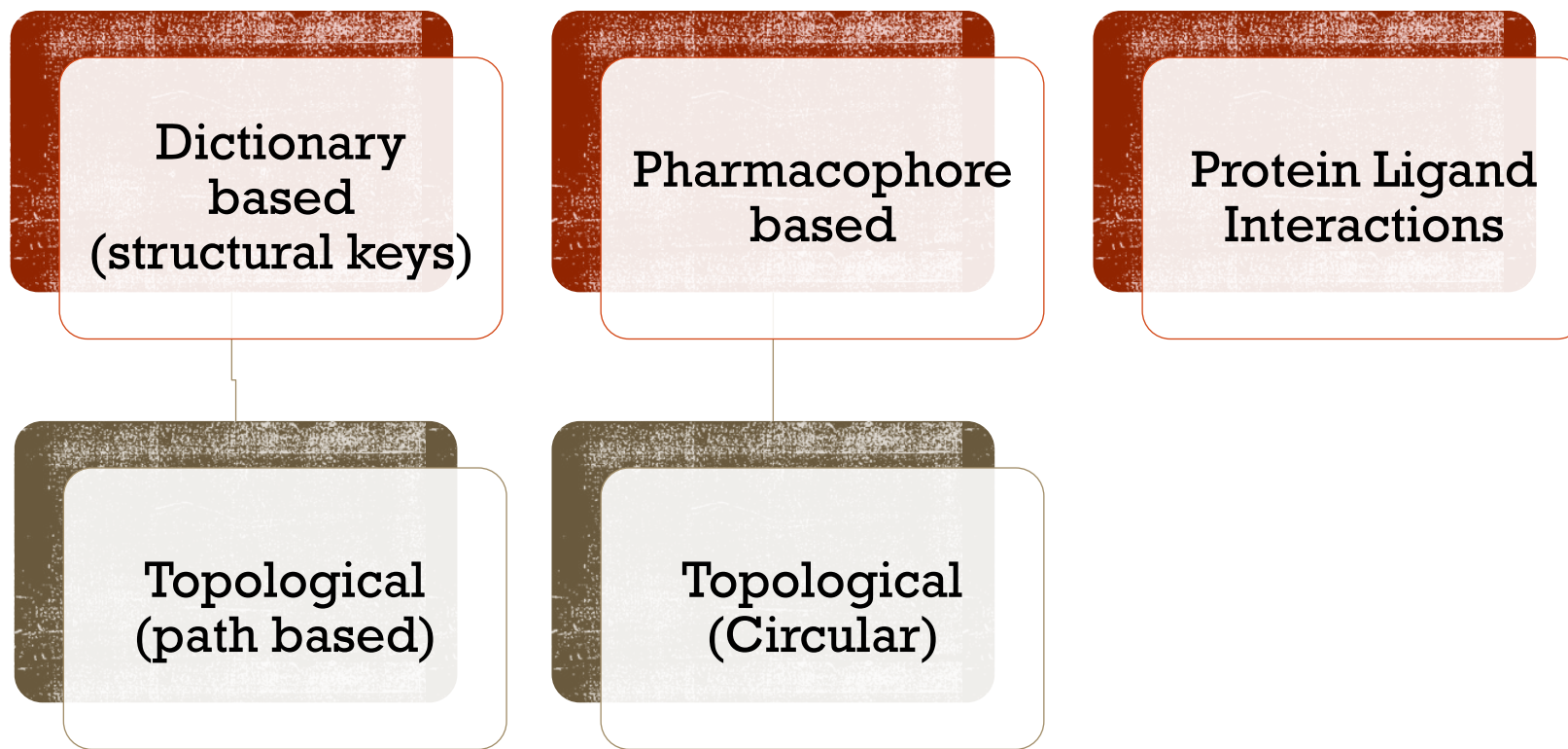
```
#calculate the full descriptors  
all_descriptors = Descriptors.CalcMolDescriptors(mol)
```

```
#calculate specific descriptors (like the Lipinski)  
mw = Descriptors.MolWt(mol)  
logp = Descriptors.MolLogP(mol)  
hba = Descriptors.NumHAcceptors(mol)  
hbd = Descriptors.NumHDonors(mol)
```



Open-Source Cheminformatics
and Machine Learning

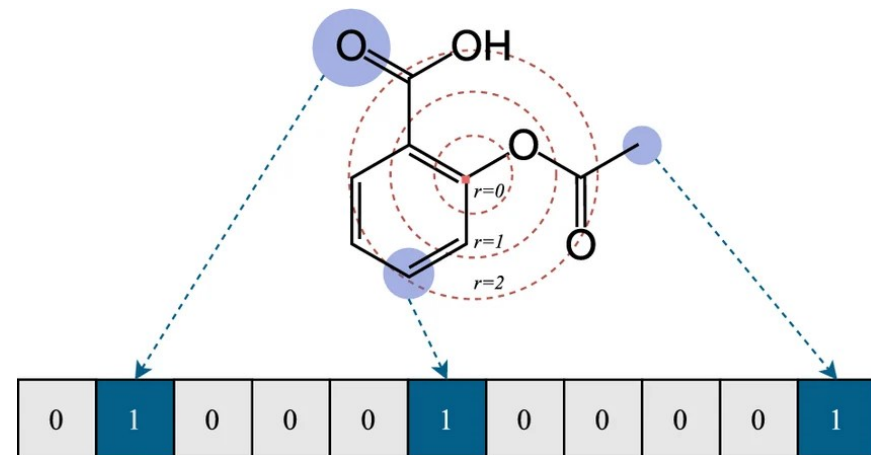
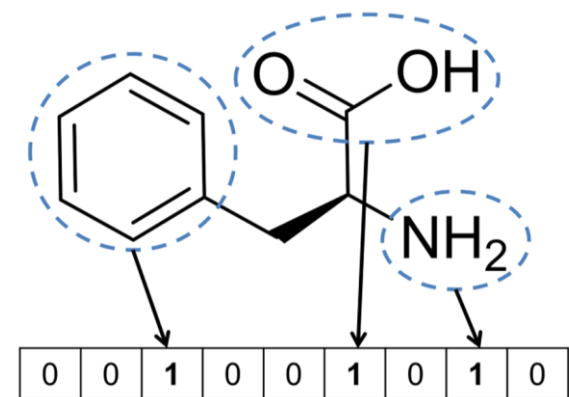
MOLECULAR FINGERPRINTS



Molecular Fingerprints

Category	Key Examples	Bit Length	Best Use Case	RDKit Function
Dictionary-based (Structural Keys)	MACCS	166	Substructure searching baseline comparisons	<code>MACCSkeys.GenMACCSKeys()</code>
Circular (Topological)	ECFP (Extended Connectivity)	Variable (2048 typical)	General similarity ML QSAR	<code>AllChem.GetMorganFingerprint()</code>
Circular (Topological)	FCFP (Functional-Class)	Variable	Binding similarity scaffold hopping	<code>AllChem.GetMorganFingerprint()</code> (<code>useFeatures=True</code>)
Circular (Topological)	Molprint2D	Variable	Atom environment similarity	Via <code>molprint2d</code> package
Topological (Path-based)	Atom Pairs (AP)	Variable	Topological similarity diversity	<code>rdkit.Chem.AtomPairs.Pairs.GetAtomPairFingerprint()</code>
Topological (Path-based)	Topological Torsions (TT)	Variable	Conformational similarity	<code>rdkit.Chem.AtomPairs.Torsions.GetTopologicalTorsionFingerprint()</code>
Topological (Path-based)	Avalon	512	Fast similarity searching	<code>AvalonTools.GetAvalonFP()</code>
Pharmacophore	PharmPrint (3-point 4-point)	Variable	3D similarity, scaffold hopping	Via <code>Pharm2D</code> or external tools

- Representations of molecules based on substructure, connectivity, graph or 3D space
- Some are count based (scalar) while others are binary (0s and 1s)



Descriptor Based Screening (Lipinski, Verbers etc.)

Lipinski's Rule (R05)



Most **drug-like molecules** in the markets in the early 2000s obeyed this rule

How do you reduce the size of your compounds based on certain features?

The Rule of 3 (RO3)

- $MW \leq 300 \text{ g/mol}$
- $CLogP \leq 3$
- Hydrogen bond acceptors ≤ 3
- Hydrogen bond donors ≤ 3
- Rotatable bonds ≤ 3
- $TPSA \leq 60$
- More important for **fragment-based discovery**

Verbers Rule

- Number of Rotatable bonds ≤ 10
- Topological Polar Surface Area (TPSA) ≤ 60
- Not standalone filter; **usually in combination with another rules**

Physicochemical Descriptors

```
from rdkit import Chem
from rdkit.Chem import Descriptors, Lipinski

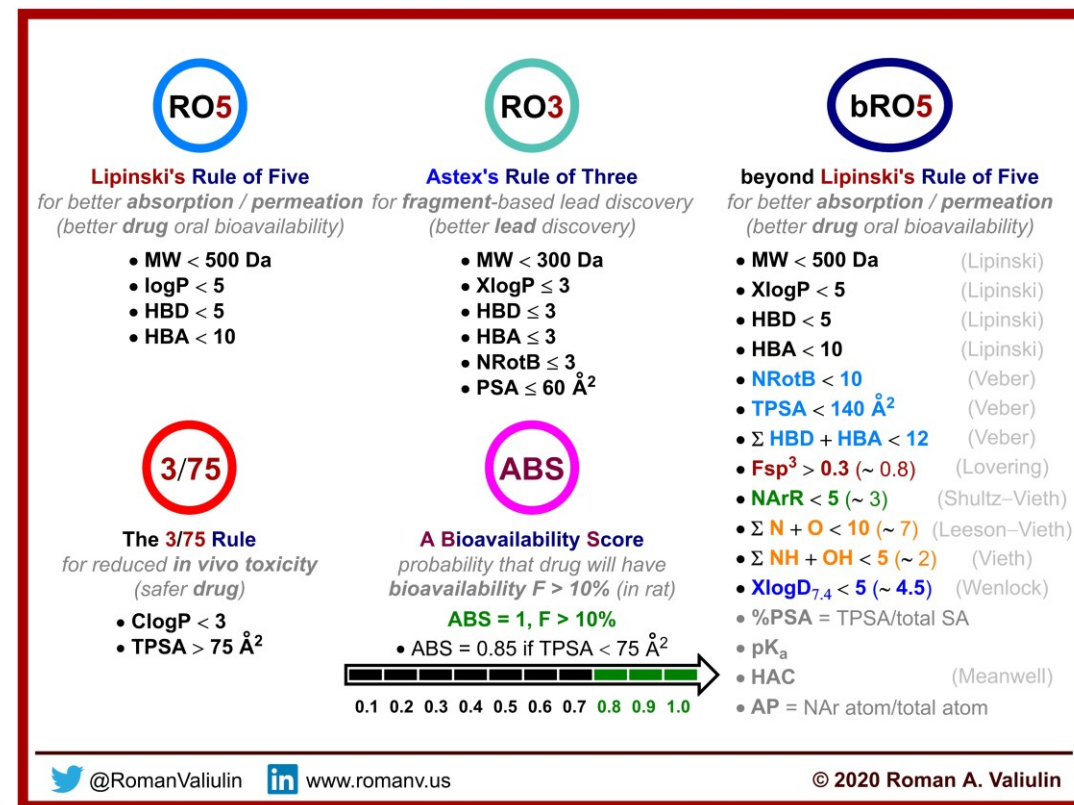
def lipinski(smiles):
    mol = Chem.MolFromSmiles(smiles)
    if not mol:
        return False
    return (Descriptors.MolWt(mol) <= 500 and
            Descriptors.MolLogP(mol) <= 5 and
            Lipinski.NumHAcceptors(mol) <= 10 and
            Lipinski.NumHDonors(mol) <= 5)

# apply to a smile list or read from a csv file
smiles_list = ["CC(=O)Oc1ccccc1C(=O)O", "CCCCO", "c1ccccc1", "CCCCNCCNCCCCCCCC"]

passed = [s for s in smiles_list if lipinski(s)]
```

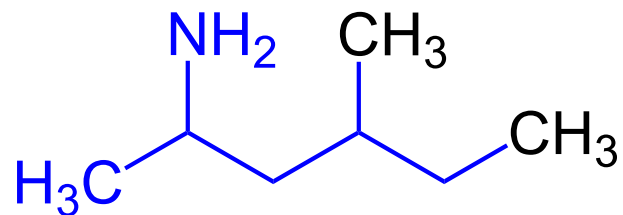
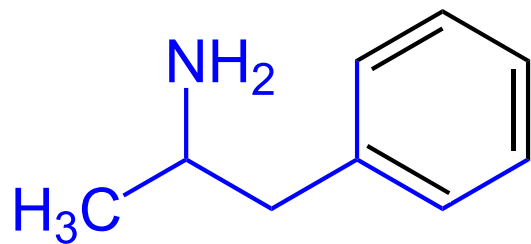


- Necessary filter in library designs and most machine learning workflows
- Does not apply for most natural products and macrocycles or peptide drugs



<https://cheminfographic.wordpress.com/2020/03/21/medchem-mnemonic-rules-lipinkis-ro5-ro3-bro5-3-75-rule/>

Chemical Similarity



How similar are these two molecules?

Molecular representations

Similarity

Scoring

Uses fingerprints to represent the molecule

- **ECPS**
- **FCPS**
- **Morgan**
- **SMIfp**

Quantify the degree of resemblance between two such representations

- **Cosine**
- **Dice**
- **Euclidean**
- **Manhattan**
- **Tanimoto**
- **Tversky**

Tanimoto Similarity

You can form a matrix of similarity indices for all pairs of molecules in your dataset and then filter at a threshold

$$T(a,b) = \frac{N_c}{N_a + N_b - N_c}$$

```
from rdkit import Chem
from rdkit.Chem import AllChem, DataStructs

def tanimoto(smiles1, smiles2):
    mol1 = Chem.MolFromSmiles(smiles1)
    mol2 = Chem.MolFromSmiles(smiles2)
    if not mol1 or not mol2:
        return None
    fp1 = AllChem.GetMorganFingerprintAsBitVect(mol1, radius=2, nBits=2048)
    fp2 = AllChem.GetMorganFingerprintAsBitVect(mol2, radius=2, nBits=2048)
    return DataStructs.TanimotoSimilarity(fp1, fp2)

s1 = "CC(=O)Oc1ccccc1C(=O)O" # Aspirin
s2 = "CC(C)CC1=CC=C(C=C1)C(C)C(=O)O" # Ibuprofen

sim = tanimoto(s1, s2)
print(f"Tanimoto similarity: {sim:.3f}")
```

Tanimoto Coefficient

- Tanimoto Coefficient of similarity for Molecules A and B:

$$S_{AB} = \frac{c}{a + b - c}$$

a = bits set to 1 in A, b = bits set to 1 in B, c = number of 1 bits common to both

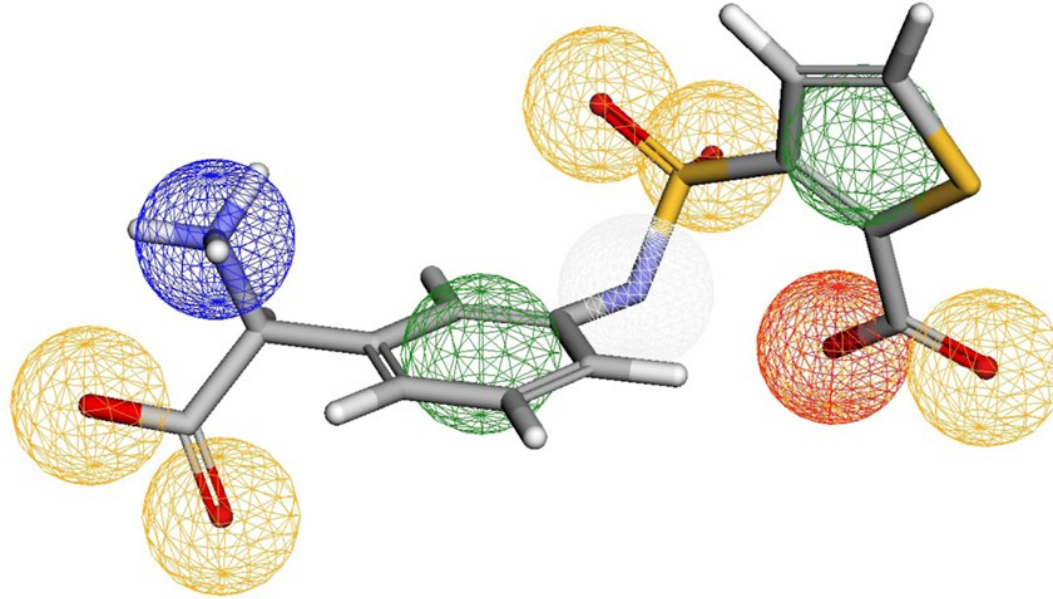
Range is 0 to 1.

Value of 1 does not mean the molecules are identical.

A	<table><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr></table>	1	0	1	1	0	1	A = 4
1	0	1	1	0	1			
B	<table><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td></tr></table>	1	1	0	1	0	0	B = 3
1	1	0	1	0	0			
A ∧ B	<table><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr></table>	1	0	0	1	0	0	A ∧ B = 2
1	0	0	1	0	0			
A ∨ B	<table><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td></tr></table>	1	1	1	1	0	1	A ∨ B = 5
1	1	1	1	0	1			

$$S_T(A, B) = \frac{2}{5}$$

Pharmacophore Screening



Feature Key:

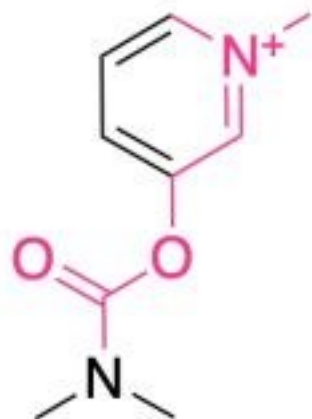
● Hydrogen Acceptor	○ Hydrogen Donor
● Hydrophobic	● Negative Ion
● Aromatic	● Positive Ion

- **A 3D template of how a ligand binds to a receptor**
- **Describes a spatial representation**
- **Describes constraints (bonds, angles and distances) that enforces such geometry in the active site**

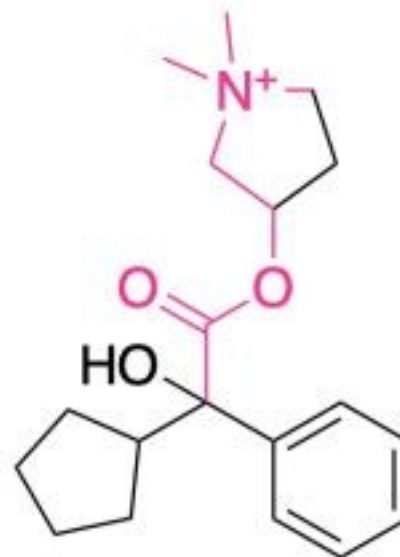
Pharmacophore Screening



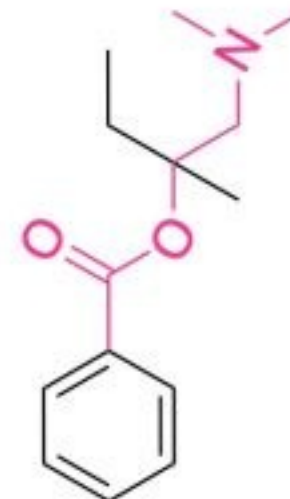
acetylcholine



pyridostigmine iodide (#1)



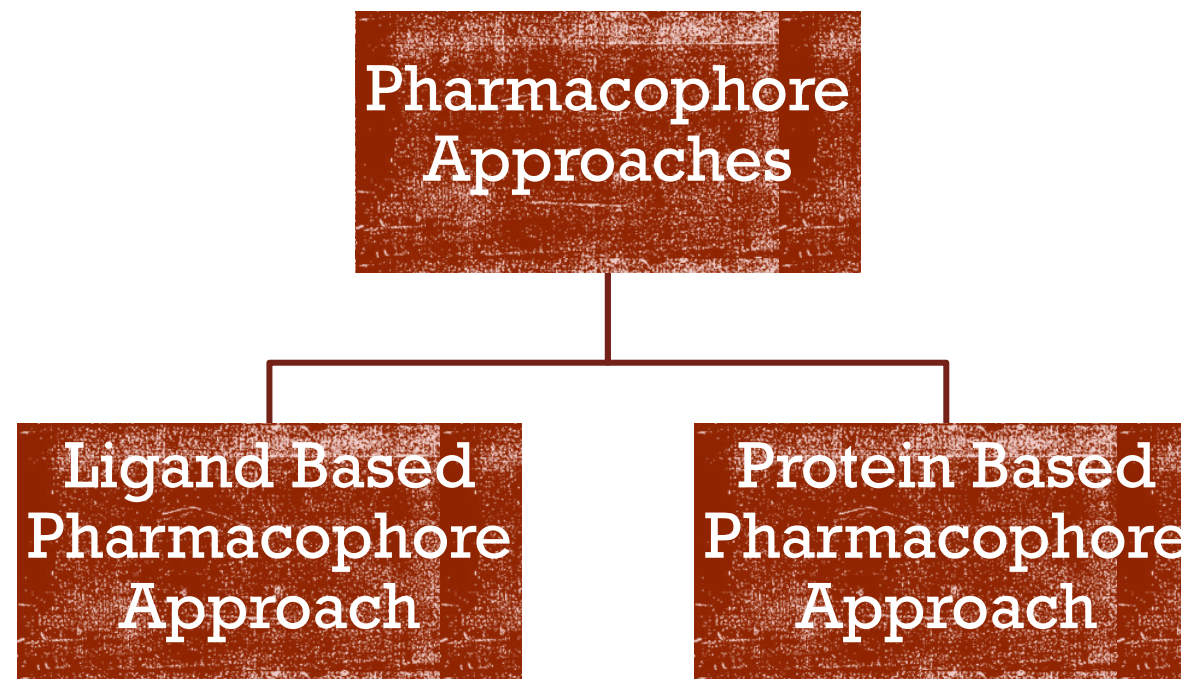
glycopyrrolate bromide (#3)



amylocaine #14

- **For acetylcholinesterase (AChE) inhibitors, these constraints are important**
- **A charged quaternary amine separated from a carboxylic group by 2-3 carbon atoms**
- **We can enforce these in search of new AChE inhibitors**

Pharmacophore Screening



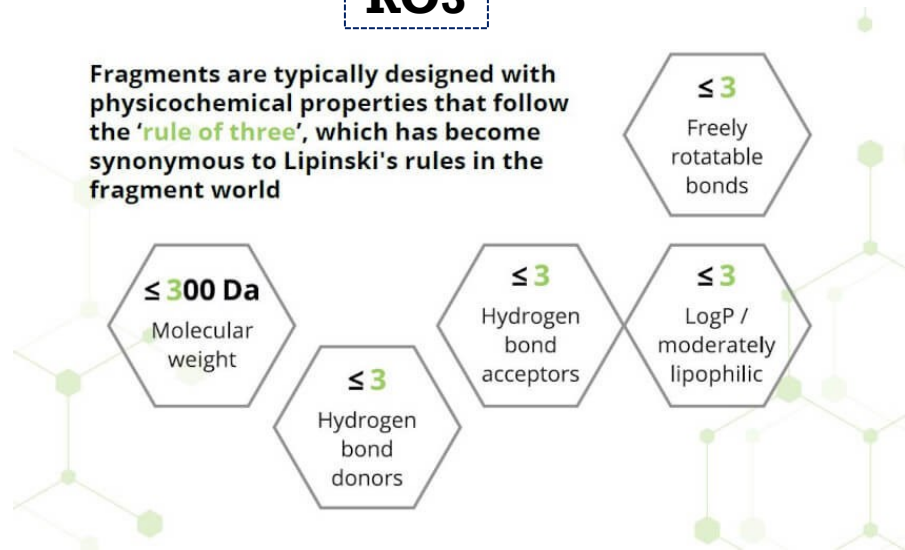
- Used from **known binders** of a given target
- **Useful in scaffold hopping**
- Ligands are aligned and then the features of pharmacophoric elements extracted

- Used in cases where the known ligands are not available
- Map out putative **sub pockets** and **excluded volumes** to define features
- Best when combined with ligand-based approach

Fragment Based Screening

RO3

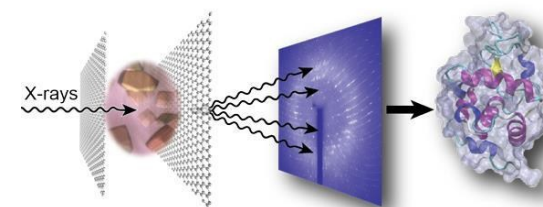
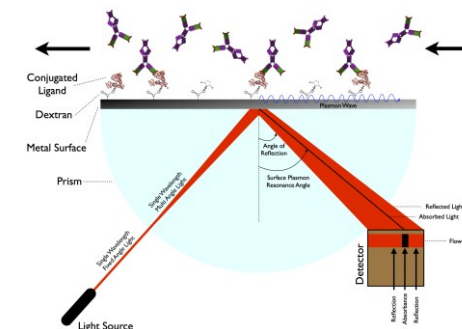
Fragments are typically designed with physicochemical properties that follow the 'rule of three', which has become synonymous to Lipinski's rules in the fragment world



Reconstruct the molecule from fragments like building from legos



Compounds are first filtered using this rule from general library or databases



Confirm the activity of the fragments from NMR, X-ray or SPR etc.

EXTRA RESOURCES

- <https://openbabel.org/docs/Fingerprints/intro.html>
- https://real.mtak.hu/72398/1/CMCIII_Chapter3_14a.pdf
- <https://pharmacelera.com/blog/science/fragment-based-drug-design-and-field-based-technology/>
- Weijun Xu and Congbao Kang Journal of Medicinal Chemistry 2025 68 (5), 5000-5004 DOI: 10.1021/acs.jmedchem.5c00424